003 Final Sorusu

a-) 5 köşeli poligon için 2. ve 3. dereceden B-spline eğrisinin düğüm vektörünü bulunuz.

$n = 5-1 = 4$

$k = 2$

$t_{max} = n-k+2$

$= 4-2+2$

$= 4$

[0 0 1 2 3 4 4]

$k = 3$

$t_{max} = n-k+2$

$= 4-3+2$

$= 3$

[0 0 0 1 2 3 3 3]

b-) 6 köşeli poligon için 3. dereceden ve 1 noktada çakışma olmak üzere B-spline eğrisinin düğüm vektörünü bulun.

$n = (6-1)-1$

$= 4$

$K = 3$

$t_{max} = n-k+2$

$= 4-3+2$

$= 3$

[0 0 0 1 1 2 3 3 3]

$n = \frac{6-3+3}{3}$

SORU B_3 ve B_4'ü elde etme

$P(t) = B_1 + B_2 t + B_3 t^2 + B_4 t^3$

$P(0) = B_1 = P_1$, $P'(0) = B_2 = P_1'$

$P(t_2) = B_1 + B_2 t_2 + B_3 t_2^2 + B_4 t_2^3 = P_2$

$P'(t_2) = B_2 + 2B_3 t_2 + 3B_4 t_2^2 = P_2'$

$-2 / P_2 = P_1 + P_1' t_2 + B_3 t_2^2 + B_4 t_2^3$

$t_2 / P_2' = P_1' + 2B_3 t_2 + 3B_4 t_2^2$

$-2P_2 = -2P_1 - 2P_1' t_2 - 2B_3 t_2^2 - 2B_4 t_2^3$

$P_2' t_2 = P_1' t_2 + 2B_3 t_2^2 + 3B_4 t_2^3$

$-2P_2 + P_2' t_2 = -2P_1 - P_1' t_2 + B_4 t_2^3$

$\frac{2(P_1-P_2)}{t_2^3} + \frac{P_1' t_2}{t_2^3} + \frac{P_2' t_2}{t_2^3} = \frac{B_4 t_2^3}{t_2^3}$

$$\frac{2(P_1-P_2)}{t_2^3} + \frac{P_1'}{t_2^2} + \frac{P_2'}{t_2^2} = B_4$$

$-3 / P_2 = P_1 + P_1' t_2 + B_3 t_2^2 + B_4 t_2^3$

$t_2 / P_2' = P_1' + 2B_3 t_2 + 3B_4 t_2^2$

$-3P_2 = -3P_1 - 3P_1' t_2 - 3B_3 t_2^2 - 3B_4 t_2^3$

$P_2' t_2 = P_1' t_2 + 2B_3 t_2^2 + 3B_4 t_2^3$

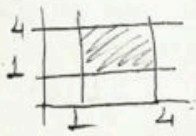
$-3P_2 + P_2' t_2 = -3P_1 - 2P_1' t_2 - B_3 t_2^2$

$3(P_1-P_2) + 2P_1' t_2 + P_2' t_2 = -B_3 t_2^2$

$\frac{3(P_2-P_1)}{t_2^2} - \frac{2P_1' t_2}{t_2^2} - \frac{P_2' t_2}{t_2^2} = \frac{B_3 t_2^2}{t_2^2}$

$$\frac{3(P_2-P_1)}{t_2^2} - \frac{2P_1'}{t_2} - \frac{P_2'}{t_2} = B_3$$

SORU



$u=1 \quad 1 \le w \le 4$
$u=4 \quad 1 \le w \le 4$
$w=1 \quad 1 \le u \le 4$
$w=4 \quad 1 \le u \le 4$

$u = \frac{y^2}{x}$, $w = \frac{x^2}{y}$

$\Downarrow$ $\Downarrow$

$x = \frac{y^2}{u}$ $y = \frac{x^2}{w}$

$x = \frac{y^2}{u} \Rightarrow x = \frac{x^4}{w^2 \cdot u} \Rightarrow \boxed{x^3 = w^2 \cdot u}$

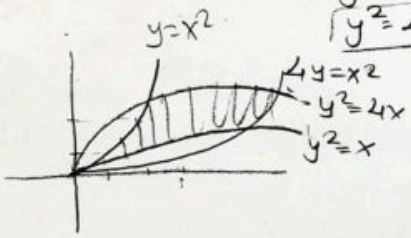
$y = \frac{x^2}{w} \Rightarrow y = \frac{y^4}{u^2 w} \Rightarrow \boxed{y^3 = u^2 \cdot w}$

u = 1 için

$x^3 = w^2$
$x^{3/2} = w$
$y^3 = w$
$y^3 = x^{3/2}$
$y = \sqrt{x} \Rightarrow \boxed{y^2 = x}$

u = 4 için

$x^3 = w^2 \cdot 4$
$w^2 = \frac{x^3}{4}$
$(y^3)^2 = (16 \cdot w)^2$
$y^6 = 16^2 \cdot \frac{x^3}{4}$
$y^6 = 64 x^3$
$\boxed{y^2 = 4x}$



w = 1 için

$x^3 = u$
$y^3 = x^6$
$\boxed{y = x^2}$

w = 4 için

$x^3 = 16u$
$u^2 = \frac{x^6}{16^2}$
$y^3 = \frac{x^6}{16^2} \cdot 4$
$\sqrt[3]{y^3} = \sqrt[3]{\frac{x^6}{64}}$
$y = \frac{x^2}{4} \Rightarrow \boxed{4y = x^2}$

